

# Q: What is the theoretical SNR of an ideal 10-bitADC?

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## A AI Model: Gemma4

**Q: What is the theoretical SNR of an ideal 10-bitADC?**

**Time: (1m 40.08s)**

**A:**

The theoretical Signal-to-Noise Ratio (SNR) of an ideal  $N$ -bit ADC, assuming the signal input covers the full-scale range ( $V_{FS}$ ) and is limited solely by quantization noise, is given by  $\text{SNR}_{\text{ideal}} \approx 6.02N + 1.76$  dB. This limit is reached because the minimum average power of the quantization noise ( $\sigma_q^2$ ) is determined by the step size  $\Delta = V_{FS}/2^N$ , resulting in a root mean square (RMS) noise voltage  $\sigma_q \approx \Delta/\sqrt{12}$ , while assuming a full-scale sine wave input gives  $V_{\text{signal, RMS}} = V_{FS}/\sqrt{2}$ . For an ideal 10-bit ADC ( $N = 10$ ), the calculated theoretical SNR is:  $\text{SNR}_{\text{theoretical}} \approx 6.02(10) + 1.76$  dB = 61.96 dB This value represents the absolute performance ceiling; in reality, this ideal SNR must be degraded by practical non-idealities such as thermal noise from amplifiers, input referred jitter ( $\sigma_t$ ), and finite comparator offset/gain bandwidth products (GBW).).

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## B AI Model: Llama3.1

**Q: What is the theoretical SNR of an ideal 10-bitADC?**

**Time: (29.64s)**

**A:**

The Signal-to-Noise Ratio (SNR) of an Analog-to-Digital Converter (ADC) can be calculated using the formula:

$$\text{SNR (dB)} = 6.02N + 1.76 \text{ dB}$$

where  $N$  is the number of bits.

For an ideal 10-bit ADC, we substitute  $N = 10$  into the equation:

$$\text{SNR (dB)} = 6.02(10) + 1.76 = 60.2 + 1.76 = 61.96 \text{ dB}$$

This result assumes an ideal ADC with no quantization noise or other non-idealities.

In decibels, this is equivalent to a SNR of  $98.32 \times 10^{(-31)}$ , however for practical purposes we can state it as:

$$\text{SNR (dB)} \approx 62 \text{ dB}$$

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## C AI Model: Qwen3.5

**Q: What is the theoretical SNR of an ideal 10-bit ADC?**

**Time: (18m 46.39s)**

**A:**

For an ideal ADC assuming negligible external circuit noise sources and no overload distortion on a full-scale sinusoidal input with peak-to-peak range  $V_{FS}$ , performance is limited solely by quantization error modeled as uniformly distributed white noise having RMS amplitude  $\sigma_q = \frac{V_{FS}}{\sqrt{12}2^N}$ ; consequently, the ratio of signal power to this floor yields an SNR derived theoretically from fundamental principles as  $\text{SNR}_{dB} = 6.02N + 1.76$ . Substituting  $N = 10$  into this expression provides a theoretical limit of approximately 62 dB, which represents the maximum achievable dynamic range provided practical non-idealities like comparator offset or sampling jitter do not dominate the noise floor in implementation.

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## D AI Model: Ministral-3

**Q: What is the theoretical SNR of an ideal 10-bit ADC?**

**Time: (1m 3.85s)**

**A:**

The **theoretical signal-to-noise ratio (SNR)** of an ideal N-bit ADC is fundamentally limited by **quantization noise**, which arises from rounding the analog input to discrete digital levels. For an ideal uniform quantizer with full-scale range  $V_{FS}$  and no other non-idealities, the SNR in decibels is given by:

$$\text{SNR}_{\text{ideal}} = 6.02N + 1.76 \quad [\text{dB}]$$

where  $N$  is the number of bits (here,  $N = 10$ ). This formula assumes:

- **Uniform quantization** (no INL/DNL errors),
- **Thermal noise and comparator offset negligible,**

- **Ideal sampling** (no jitter or aperture uncertainty),
- **Full-scale sine-wave input** with amplitude  $V_{FS}/2$ .

For  $N = 10$ , the theoretical SNR is:

$$\text{SNR} = 6.02 \times 10 + 1.76 = 61.96 \text{ dB}$$

This result stems from the **power spectral density (PSD) of quantization noise**, which is white and uniformly distributed across the Nyquist band, while the signal power scales with  $N^2$ . The extra +1.76 dB accounts for the **peak-to-RMS conversion** of a sine wave relative to the full-scale range. In practice, real ADCs deviate due to non-idealities (e.g., comparator offset, finite op-amp gain), reducing SNR below this ideal bound..

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